Cell Design Specification — VBF Power-Tool Cell (V4 Margin-fix)

Case: t3_r1_flash | Generation date: 2026-08-25 | Doc: VBF-T3R1FLASH-DS-01

1. Basic Specification

Item	Value	Source
Electrochemical system	NMC-type positive / graphite negative (Chen2020 parameterization)	base_params=Chen2020 (entry 0 meta)
Nominal capacity	3.44 Ah	1C DFN sim: cell/r2_v4_1c_dfn.json:capacity_ah=3.4376
Voltage window	2.5 – 4.2 V	parameter set (Lower/Upper voltage cut-off)
Cell dimensions	height 65 mm × width 1580 mm (wound strip) × stack 0.14 mm	parameter set (Electrode height/width), calc-energy thickness_m=1.4e-4
Shell / casing thickness	Not provided (no casing parameter in set)	—
Electrolyte formulation	high-transport LiPF6-class formulation: σ_e=2.4 S/m, D_e=3.5e-10 m^2/s, t■=0.5 (design overrides, estimate)	params_r2_v4.json (estimate, literature-adjacent)
Electrode area	0.1027 m^2	calc-energy area_m2

2. Electrode and Separator

Layer	Thickness (µm)	Porosity	Particle radius (µm)	Material / note
Positive electrode	42	0.40	0.8	NMC-type; design value (baseline 75.6 µm / 0.335 / 5.22 µm)
Negative electrode	58	0.38	1.0	graphite; design value (baseline 85.2 µm / 0.25 / 5.86 µm)
Separator	12	0.47	—	parameter set
Positive current collector	16	—	—	Al (2700 kg/m^3), parameter set
Negative current collector	12	—	—	Cu (8960 kg/m^3), parameter set

N/P ratio = negative capacity density × thickness ÷ positive capacity density × thickness

= (0.62×33133×58)/(0.60×63104×42) = **0.75** (full-range capacity-density definition; usable-window N/P is higher — cell is cathode-limited in simulation, anode margin +12% vs baseline).

3. Process Design Parameters

Parameter	Formula	Value	Unit
Positive areal density	thickness × (1–porosity) × density	42e-6×0.60×3262 = 82.2 g/m^2	g/m^2
Negative areal density	thickness × (1–porosity) × density	58e-6×0.62×1657 = 59.6 g/m^2	g/m^2
Positive compaction density	density × (1–porosity)	3262×0.60 = 1957 kg/m^3 = 1.96 g/cm^3	g/cm^3
Negative compaction density	density × (1–porosity)	1657×0.62 = 1027 kg/m^3 = 1.03 g/cm^3	g/cm^3
Electrolyte fill amount	pore volume × electrolyte density × fill factor	4.57 mL × 1.2 g/cm^3 × 1.0 = 5.5 g	g (literature density 1.2 g/cm^3, annotated)
Formation recommendation	0.1C CC to 4.2 V, 25 °C, 2 cycles	design recommended value; actual production-line value requires tuning	—

4. Mass Breakdown (contract caliber; electrolyte excluded)

Layer	kg/m^2	Mass (g)
Positive electrode	0.08220	8.44
Negative electrode	0.05959	6.12
Positive CC (Al)	0.04320	4.44
Negative CC (Cu)	0.10752	11.04
Separator	0.00253	0.26
Total (cell mass for energy density)		**30.30**
Electrolyte (informational)		5.48 (estimate)

Source: calc-energy cell/r2_v4_energy.json (layer_kg_m2 × area 0.1027 m^2); electrolyte excluded from contract mass (parameter set lacks density) — annotated.

5. Performance Verification (vs entry-0 criteria)

Metric	Threshold	Value	Verdict	Source
capacity_ah	≥ 2.0 Ah	3.438 Ah	✓	r2_v4_1c_dfn.json:capacity_ah
retention_5c	≥ 0.95	0.987	✓	r2_v4_derived.json:retention_5c (3.393/3.438)
power_density_w_kg	≥ 4000 W/kg	111,807 W/kg	✓	r2_v4_energy.json:power_density_w_kg
T_max_K (4C charge @45 °C)	≤ 333.15 K (60 °C)	321.94 K (48.8 °C)	✓	r2_v4_4c_dfn.json:T_max_K
plated (4C charge)	false	false (anode min +22.6 mV)	✓	r2_v4_4c_dfn.json:anode_potential_v

6. Design Notes (rationale from evaluate log)

• R0 baseline (Chen2020 unmodified): capacity 4.95 Ah ✓, power density 22,383 W/kg ✓, but retention_5c 0.087 ✗, plated=true ✗, T_max 354.3 K ✗. Root cause: positive solid diffusivity 4e-15 m^2/s → particle diffusion time constant ~6.8 ks at r=5.22 µm vs 720 s 5C window (excluded lever: solid diffusivity; allowed lever: particle radius).

• R1: thin electrodes (42/52 µm), porosity 0.40/0.35, particles 1.0/1.5 µm, σ_e 2.0 S/m, t■ 0.4, h=100: retention 0.983 ✓, T_max 323.3 K ✓, power ✓, capacity 3.08 ✓; plated=true marginal (–3.9 mV at final charge instant only).

• R2 (final, V4): thicker negative (58 µm, ε 0.38, r 1.0 µm) raises end-of-charge anode potential floor; r_pos 0.8 µm improves charge acceptance; σ_e 2.4 S/m, t■ 0.5, D_e 3.5e-10: plated=false (anode min +22.6 mV), retention 0.987, T_max 321.9 K — all criteria pass.

• Conservative alternative V5 (anode-geometry fix only, V1-level transport) also passes all criteria (plated false, +16.0 mV).